

C.4 Seizure Alerts from EEG Readings

In this experiment, we use the real-world CHB-MIT Scalp EEG Database [16], a collection of EEG recordings of 22 pediatric patients with intractable seizures, to evaluate the ability of our suite of test martingales to robustly detect seizure in patients from EEG data. The onsets and ends of 182 seizures are annotated in the dataset.

Figure 12 shows the EEG measurements for one of the patients (https://physionet.org/files/chbmit/1.0.0/chb01/chb01_03.edf), where the red vertical line shows the time when a seizure started. There are 256 EEG readings per second.

We trained a CTW of depth 4 over the sequence and ran a marginal, conditional, and conformal martingales over the generated p-values. Figure 13 shows the results of the test martingales over twenty random seeds. As predicted by the theory, the marginal martingale (with power gambler betting function) can successfully isolate the distribution shift soon after the onset of seizure. The conditional martingale (with autocorrelation betting function) generated some false alarms, but this can be moderated using the conformal martingale on the same autocorrelation betting function.

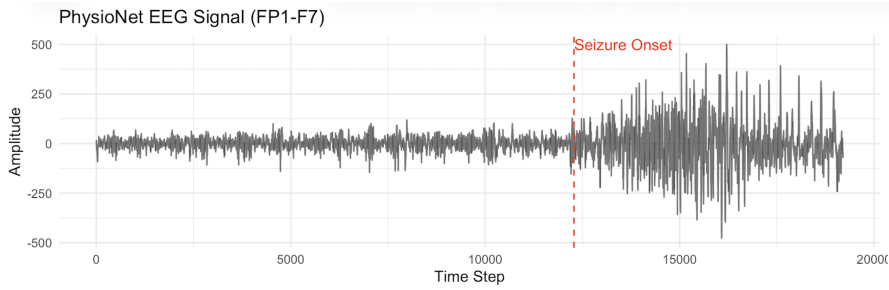


Figure 12: EEG reading for a patient who suffered a seizure midway through the period.

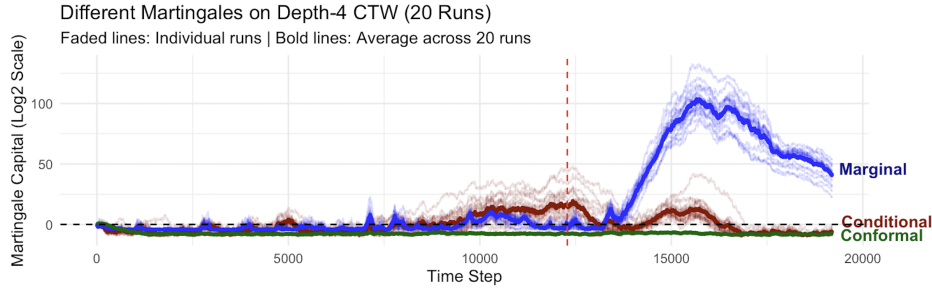


Figure 13: The martingale scores for a suite of test martingales on the data.

C.5 Transformer Martingale

Figure 14 shows the behaviour of a transformer martingale when the underlying CTW model is well-specified.

Each data point in the data sequence is generated by taking the XOR of 3 data points in the preceding window of size 5. A CTW of depth 5 is used to process the data sequence and, since the CTW model is well-specified, it correctly predicts each data point in the sequence with near-perfect accuracy.

A transformer martingale is used to process the p-values generated by the CTW model. The transformer architecture has a single-head self-attention layer plus a multi-layer perceptron for final binary prediction, with positional encoding for the sequence of p-values. The context window is set at 5, similar to that for the CTW depth.

As expected, the martingale score goes down consistently over time.

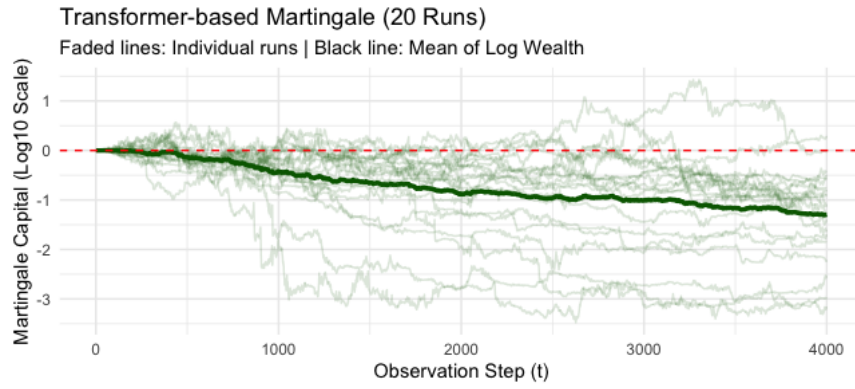


Figure 14: The behavior of the transformer martingale when the underlying CTW model is well-specified.